

RANK-ONE NONSEMISIMPLE EXTENSIONS OF CYCLIC FUSION CATEGORIES

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ABSTRACT. Gabriel’s theorem encodes the abelian-categorical structure of a hereditary finite-type algebra entirely in a directed graph. The tensor-categorical generalization of this principle asks for a quiver datum that generates a monoidal category, its module categories, and the explicit action formula as output rather than prescribing a fixed ambient tensor environment. We develop this program for rank-one quivers.

The input is a *rank-one convolution source datum* (G, a, χ) : an abelian group G , a shift $a \in G$, and a character $\chi: G \rightarrow k^\times$. Replacing the diagonal (Herschend) tensor product on quiver representations with group-algebra convolution on the support forces the arrow map to satisfy a Leibniz rule, and monoidal coherence forces χ to be a group homomorphism. The tensor category $\mathcal{C}(G, a, \chi)$, together with its module categories and action formulas, is recovered from this combinatorial datum.

On the tensor-category side, we prove two recognition theorems: the line datum $G = \mathbb{Z}$, $a = 1$, $\chi(i) = q^i$ yields a category equivalent to $\mathbb{Z}$ -weight modules over the quantum Borel algebra B_q , and at a primitive ℓ -th root of unity the nilpotent cyclic quotient $\mathcal{C}_\ell^{(\ell)}(q)$ is equivalent to $\text{Rep}(T_\ell(q))$ for the Taft algebra $T_\ell(q)$, which is rigid and pivotal. The hierarchy $U(\mathfrak{b}) \rightsquigarrow B_q \rightsquigarrow T_\ell(q)$ is selected by the source datum, not assumed in advance.

On the module-category side, we classify all finite indecomposable exact left, right, and bimodule categories of Leibniz type over the cyclic quotient categories $\mathcal{C}_d^{(e)}(\zeta)$ completely and explicitly: they are parametrized by a divisor $c \mid d$ and a scalar $\xi \in k$, with the action written down as a Leibniz rule. Where the Etingof–Ostrik and Andruskiewitsch–Mombelli classifications proceed via algebra objects or comodule algebras internal to the tensor category, ours proceeds via combinatorial divisibility data external to it.

We also characterize descent: the pushforward π_* along a group quotient $\pi: G \rightarrow H$ is strong monoidal precisely when χ factors through H . For $\pi: \mathbb{Z} \rightarrow \mathbb{Z}/\ell\mathbb{Z}$ this recovers the root-of-unity hypothesis as a descent condition. The gcd–lcm pullback square controls all descent bimodule categories between cyclic Leibniz categories. The monoidal coproduct structure on the path algebra, the Frobenius structure, and the Picard group of the nilpotent quotient categories are determined, and the convolution branch is separated from the diagonal branch via support behavior.

1. INTRODUCTION

Gabriel’s theorem [1] established that every hereditary finite-type algebra is Morita equivalent to the path algebra of a Dynkin quiver, and that the indecomposable modules correspond bijectively to the positive roots of the underlying root system [25, 4]. The force of this result is not merely classificatory. It says that the action of an algebra on its modules — the full abelian-categorical structure, not just a list of indecomposables — can be read off from a directed graph: the path algebra is the acting object, an indecomposable representation is an object of the acted-on category, and every structural fact about the action (extensions, Auslander–Reiten sequences, projective resolutions) becomes a combinatorial statement about paths and relations in the quiver.

The action of an algebra on a module is the classical setting; the natural generalization is the action of a tensor category on a module category. The most systematic treatment of tensor categories acting on module categories is due to Etingof and Ostrik [5, 6], who developed the theory of exact module categories and showed that indecomposable module categories are classified by algebra objects internal to the tensor category, up to Morita equivalence. In the case of finite tensor categories, they also show that a finite tensor category equipped with an exact fiber functor is monoidally equivalent to the representation category of a finite-dimensional Hopf algebra. Andruskiewitsch and Mombelli [7] gave a dual formulation of this classification in terms of left comodule algebras, whose coaction encodes the same module-category data in coalgebraic form.

A natural development of Gabriel’s theorem asks how quiver path algebras behave when the classical module setting is replaced by a tensor-categorical one. Etingof, Kinser, and Walton [8] embed path algebras and tensor algebras into finite tensor categories and classify their actions in terms of module-category data. Kinser and Walton [9] classify actions of Taft algebras on path algebras of finite Schurian quivers, and Kinser and Oswald [10] extend this to quantum Borels, generalized Taft algebras, and small quantum groups acting on path algebras. In each of these works the quiver continues to describe the object being acted upon — a path algebra — under the action of a fixed, externally specified quantum symmetry.

A different role for the quiver arises when the graph data is attached not to the acted-upon algebra but to the acting symmetry itself. This perspective originates with the Hopf quivers of Cibils and Rosso [19], who showed that the path coalgebra of a quiver carries a graded Hopf algebra structure precisely when the vertices form a group and the arrows carry compatible Hopf-bimodule data. Here the combinatorial data describes the acting Hopf algebra rather than its representations: the quantum group itself is the primary output, and the monoidal structure of its module categories is not the object of study. Related constructions place bialgebra structures on path

coalgebras and obtain Gabriel-type theorems for pointed bialgebras [20], but again at the level of the algebra rather than a prescribed monoidal structure on quiver representations.

A parallel line of inquiry asks when the representation category of a quiver itself carries a monoidal structure. Herschend [11, 12] studied a pointwise, diagonal tensor product on $\text{Rep}(Q)$ in which the arrow maps act in grouplike fashion. More recent work situates quiver representations within fusion-categorical environments. Schauermann [23] constructs rigid monoidal structures on categories of quiver modules via actions of monoidal categories on quivers, obtaining a class of preprojective algebras whose modules carry a compatible fusion product. Heng [21] and Elias–Heng [22] introduce Coxeter quivers and their generalization, fusion quivers, taking representations valued in a module category over an ambient fusion category and proving a Gabriel-type classification in which finite-type diagrams and indecomposables correspond to Coxeter–Dynkin data and positive roots. Heng and Stroński [24] develop $\mathcal{C}$ -species, an analog of species and quivers for algebras internal to a fusion category, and show that every such algebra is Morita equivalent to an admissible quotient of a $\mathcal{C}$ -species path algebra, hereditary exactly when no quotient is required. In each case the quiver, species, or representation lives inside a tensor or fusion category that is supplied in advance.

Our framework occupies the remaining position: we work at the level of tensor categories, as in the fusion-categorical constructions above, but the quiver data is the generating, acting source rather than an object placed inside a pre-existing tensor environment. The rank-one convolution source datum (G, a, χ) is the input, and the tensor category, its module categories, and the explicit action formula are recovered as output. This shifts the organizing principle from a pointwise, Cartesian-like monoidal structure to a group-algebra-like convolution structure on the support, and it is this convolution that forces the arrow map to obey a Leibniz rule.

In Herschend’s setting, the monoidal structure on $\text{Rep}(Q)$ is pointwise: the tensor product at a vertex x sees only the fibers at x ,

$$(V \otimes W)(x) = V(x) \otimes W(x), \quad (V \otimes W)(\alpha) = V(\alpha) \otimes W(\alpha),$$

and the arrow acts diagonally, or grouplike. This is the natural choice when the vertex set has no binary operation: the simples $S(x)$ satisfy $S(x) \otimes S(y) = 0$ for $x \neq y$, so the support of a tensor product is the intersection of supports. Essentially, our approach shifts the paradigm from a Cartesian-like monoidal structure to a group-algebra-like convolution structure.

If the vertex set is an abelian group G , a genuinely different possibility opens. One can ask for the support of $M \otimes N$ at $\lambda \in G$ to receive contributions from all pairs

(α, β) with $\alpha + \beta = \lambda$:

$$(M \otimes N)_\lambda = \bigoplus_{\alpha+\beta=\lambda} M_\alpha \otimes N_\beta.$$

This is group-algebra convolution on the support, and it fundamentally alters the underlying representation theory. The simples $S(\alpha)$ now satisfy $S(\alpha) \otimes S(\beta) \cong S(\alpha + \beta)$, so the group structure on the vertex set becomes a group structure on simples. Once the support is convolutional, the arrow map is forced to change: it can no longer be grouplike, and must instead satisfy a Leibniz rule

$$\partial(m \otimes n) = \partial m \otimes n + \chi(|m|) m \otimes \partial n$$

for some function χ of the degree of m . The monoidal coherence condition forces χ to be a group homomorphism (Proposition ??), i.e., a character of G . The triple (G, a, χ) — grading group, shift, character — is the *rank-one convolution source datum*: the minimal quiver-theoretic data from which the tensor category, its module categories, and the explicit action formula can all be recovered.

We realize this recovery in two main directions. On the tensor-category side: the line case $G = \mathbb{Z}$, $a = 1$, $\chi(i) = q^i$ gives a monoidal category equivalent to the category of $\mathbb{Z}$ -weight modules over the quantum Borel algebra B_q (Theorem ??). At a primitive ℓ -th root of unity q , the nilpotent cyclic quotient is equivalent to $\text{Rep}(T_\ell(q))$ for the Taft algebra $T_\ell(q)$ (Theorem ??), which is rigid and pivotal [6]. The relevant quantum group background is in [14, 15, 16]; the Taft algebra originates in [13]. The rank-one hierarchy

$$U(\mathfrak{b}) \rightsquigarrow B_q \rightsquigarrow T_\ell(q)$$

is recovered from the source datum by imposing tensor-stability on path relations — we do not start with the Hopf algebra and then ask about quivers; we start with the quiver and recognize which Hopf algebra it selects.

On the module-category side: the classification is explicit and computable. Every finite indecomposable exact left module category of Leibniz type over the quiver-derived category $\mathcal{C}_d^{(e)}(\zeta)$ is the category $\mathcal{M}_{c,\xi}$ of $(\mathbb{Z}/c\mathbb{Z})$ -graded spaces with a degree-one operator satisfying $\delta^e = \xi \text{id}$, for admissible parameters (c, ξ) (Theorem ??). The action formula is the Leibniz rule: $\delta(m \otimes v) = \partial m \otimes v + \zeta^{|m|} m \otimes \delta v$. This makes the comparison with EO/AM precise: where their classifications proceed via algebra objects or comodule algebras, ours proceeds via a divisor $c \mid d$ and a scalar $\xi \in k$, and the action is written down explicitly. The right and bimodule cases are parallel (Theorems ??, ??).

The covering structure returns in the descent results. The convolutional counterpart of Herschend's projection formula is the descent criterion: the pushforward π_* along a quotient $\pi: G \rightarrow H$ is strong monoidal precisely when the character χ factors through H (Theorem ??). For $\pi: \mathbb{Z} \rightarrow \mathbb{Z}/\ell\mathbb{Z}$, this condition is $q^\ell = 1$:

the root-of-unity hypothesis is the descent condition, not an external Hopf-algebraic input. The descent structure also organizes the bimodule categories: the gcd-lcm square of grading groups controls all descent bimodule categories between $\mathcal{C}_a^{(e)}(\zeta)$ and $\mathcal{C}_b^{(e)}(\zeta)$, with the canonical one sitting at $h = \gcd(a, b)$ (Theorem ??).

REFERENCES

- [1] P. Gabriel, Unzerlegbare Darstellungen I, *Manuscripta Math.* **6** (1972), 71–104.
- [2] K. Bongartz and P. Gabriel, Covering spaces in representation-theory, *Invent. Math.* **65** (1982), 331–378.
- [3] M. Auslander, I. Reiten, and S. O. Smalø, *Representation Theory of Artin Algebras*, Cambridge Studies in Advanced Mathematics 36, Cambridge University Press, 1995.
- [4] I. Assem, D. Simson, and A. Skowroński, *Elements of the Representation Theory of Associative Algebras, Vol. 1*, London Mathematical Society Student Texts 65, Cambridge University Press, 2006.
- [5] P. Etingof and V. Ostrik, Finite tensor categories, *Mosc. Math. J.* **4** (2004), no. 3, 627–654.
- [6] P. Etingof, S. Gelaki, D. Nikshych, and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, Providence, RI, 2015.
- [7] N. Andruskiewitsch and M. Mombelli, On module categories over finite-dimensional Hopf algebras, *J. Algebra* **314** (2007), no. 1, 383–418.
- [8] P. Etingof, R. Kinser, and C. Walton, Tensor algebras in finite tensor categories, *Int. Math. Res. Not. IMRN* (2021), no. 24, 18529–18572.
- [9] R. Kinser and C. Walton, Actions of some pointed Hopf algebras on path algebras of quivers, *Algebra Number Theory* **10** (2016), no. 1, 117–154.
- [10] R. Kinser and D. Oswald, Hopf actions of some quantum groups on path algebras, *J. Algebra* **587** (2021), 85–119.
- [11] M. Herschend, Galois coverings and the Clebsch–Gordan problem for quiver representations, *Colloq. Math.* **109** (2007), no. 2, 193–215.
- [12] M. Herschend, Tensor products on quiver representations, *J. Pure Appl. Algebra* **212** (2008), no. 2, 452–469.
- [13] E. J. Taft, The order of the antipode of finite-dimensional Hopf algebra, *Proc. Natl. Acad. Sci. USA* **68** (1971), no. 11, 2631–2633.
- [14] G. Lusztig, *Introduction to Quantum Groups*, Progress in Mathematics 110, Birkhöhm, Boston, 1993.
- [15] V. Chari and A. Pressley, *A Guide to Quantum Groups*, Cambridge University Press, 1994.
- [16] C. Kassel, *Quantum Groups*, Graduate Texts in Mathematics 155, Springer, New York, 1995.
- [17] S. Mac Lane, Natural associativity and commutativity, *Rice University Studies* **49** (1963), no. 4, 28–46.
- [18] G. Böhm and K. Szlachányi, Hopf algebroids with bijective antipodes: axioms, integrals, and duals, *Comm. Algebra* **32** (2004), no. 11, 4433–4464.
- [19] C. Cibils and M. Rosso, Hopf quivers, *J. Algebra* **254** (2002), no. 2, 241–251.
- [20] H.-L. Huang and B. Torrecillas, Quiver bialgebras and monoidal categories, *Colloq. Math.* **131** (2013), no. 2, 287–306.
- [21] E. Heng, Coxeter quiver representations in fusion categories and Gabriel’s theorem, *Selecta Math. (N.S.)* **30** (2024), no. 4, Paper No. 67.

- [22] B. Elias and E. Heng, Classification of finite type fusion quivers, preprint, arXiv:2404.09714, 2024.
- [23] G. Schaumann, Fusion quivers, *Comm. Contemp. Math.*, published online 2025, arXiv:2307.09229.
- [24] E. Heng and M. Stroiński, Algebras in fusion categories: species and hereditary algebras, preprint, arXiv:2602.19144, 2026.
- [25] P. Gabriel and A. V. Roiter, *Representations of Finite-Dimensional Algebras*, Encyclopaedia of Mathematical Sciences, vol. 73, Springer, Berlin, 1992.